Unsupervised Learning Analysis of Breast Cancer Cell Nuclei Characteristics

1. Introduction

This study analyses a dataset of breast cancer cell nuclei characteristics, including features related to size, shape, and structural complexity. As no labels are provided, unsupervised learning methods are used to explore the underlying structure of the data.

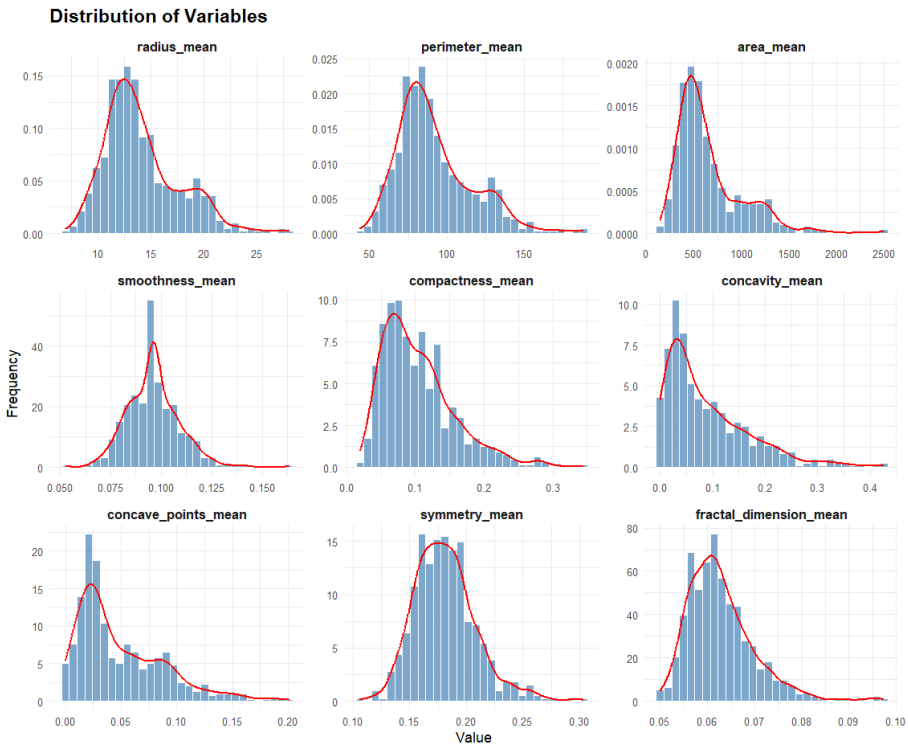
The analysis consists of three stages: exploratory data analysis and handling of missing values, dimensionality reduction using Principal Component Analysis (PCA), and clustering to identify potential groups within the data.

2. Exploratory and Missing Data Analysis

2.1 Variable Characteristics

The variables describe different characteristics of cell nuclei, including size (e.g., radius, perimeter, and area), shape irregularity (e.g., concavity and compactness), and structural complexity (e.g., smoothness, symmetry, and fractal dimension).

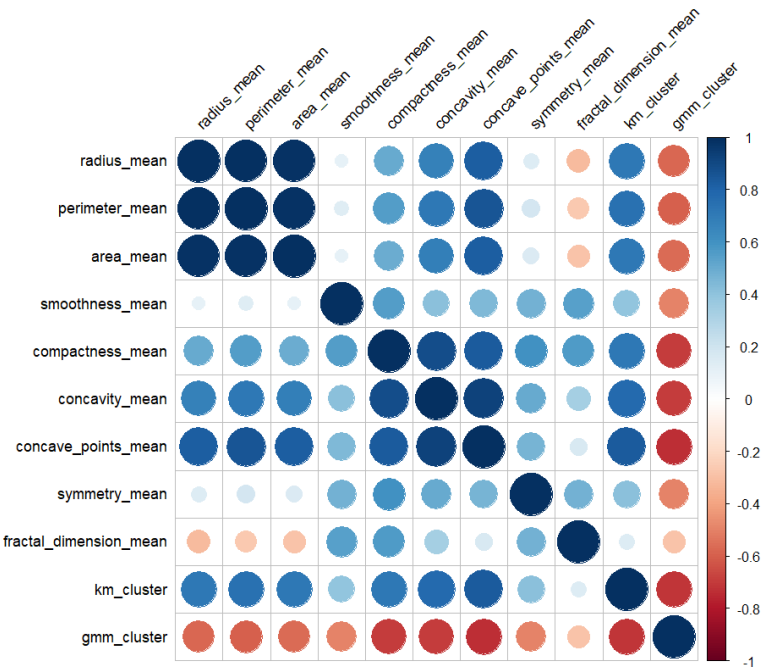
Figure 1: Histogram of variables



The distributions of the variables are shown in Figure 1. Several variables, particularly those related to size and shape (such as perimeter_mean, area_mean, and concavity_mean), exhibit right-skewed distributions, indicating the presence of larger extreme values. In contrast, variables such as smoothness_mean and symmetry_mean appear more symmetric.

2.2 Relationships Between Variables

Figure 2: Correlation matrix



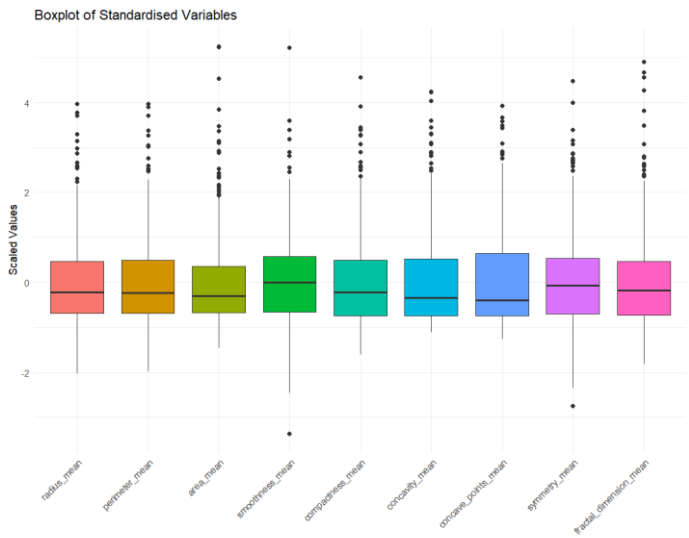
The correlation matrix (Figure 2) reveals strong positive relationships among size-related variables, including radius_mean, perimeter_mean, and area_mean. Similarly, compactness_mean, concavity_mean, and concave_points_mean are highly correlated, suggesting they capture related aspects of nuclear shape irregularity.

In contrast, variables such as smoothness_mean and symmetry_mean show weaker correlations with other features, indicating that they contribute additional but less dominant information. Interestingly, there is a weak negative correlation between fractal_dimension_mean and size variables, suggesting that it captures distinct structural characteristic not related to nuclear size.

Overall, the high correlation suggests redundancy among size-related variables, motivating the use of dimensionality reduction in the next section.

2.3 Outliers and Extremes

Figure 3: boxplots of standardised variables



Boxplots of the standardised variables (Figure 3) indicate the presence of outliers across several features, particularly those related to size and shape. These extreme values are consistent with the observed right-skewness in the distributions.

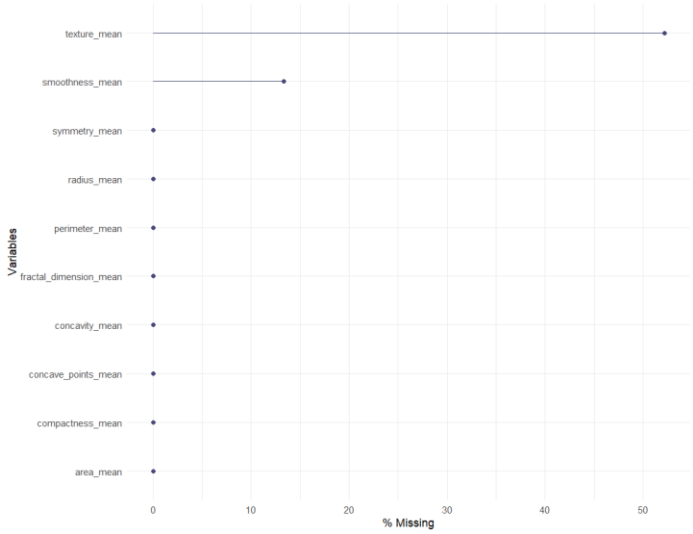
While outliers may influence distance-based methods such as K-means clustering, they are retained in this analysis as they

may represent meaningful biological variation rather than measurement errors.

2.4 Missing data

The missing data pattern is summarised in Figure 4. Approximately 50% of the values in texture_mean and 13% in smoothness_mean are missing.

Figure 4: Missing data pattern across variables



Due to the high proportion of missing values in texture_mean, this variable is removed from the analysis. For smoothness_mean, mean imputation is applied to replace missing values. This approach is simple and preserves the overall distribution of the variable, making it suitable for exploratory analysis and clustering. While mean imputation may reduce variance, this limitation is not critical in this context as the analysis focuses on structure rather than statistical inference.

2.5 Pre-processing

Before applying dimensionality reduction and clustering, the data was standardised to ensure that all variables contribute equally to the analysis. This step is necessary because the variables are measured on different scales, and distance-based methods are sensitive to scale differences.

3. <u>Dimension Reduction</u>

3.1 Choice of Method

Principal Component Analysis (PCA) was selected as the dimensionality reduction technique. The dataset consists of continuous variables with strong correlations, particularly among size-related features such as radius_mean, perimeter_mean, and area_mean. PCA is therefore appropriate as it reduces redundancy and summarises the main variation in the data.

Factor Analysis was not chosen, as the aim is not to model latent variables, and Correspondence Analysis is not suitable since the data is not categorical.

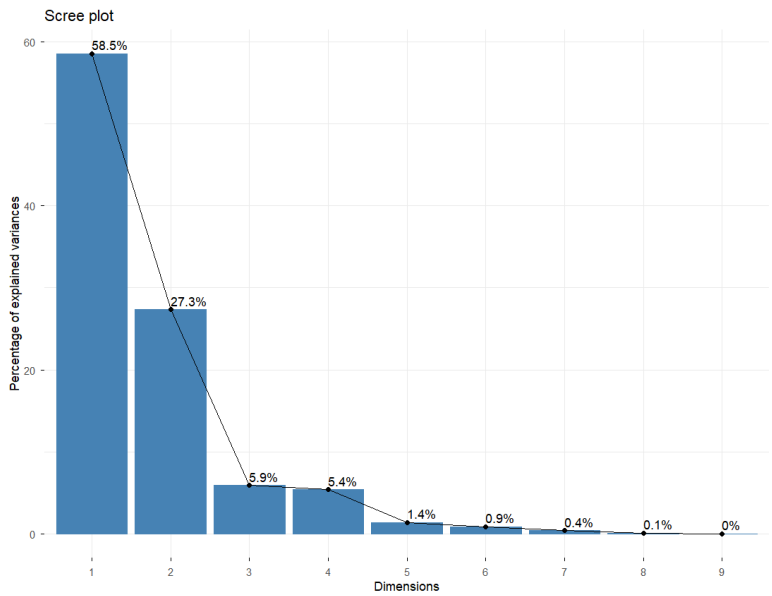
3.2 Procedure

Before applying PCA, the dataset was pre-processed by removing texture_mean due to its high proportion of missing values, imputing missing values in smoothness_mean using the mean, and standardising all remaining variables. Standardisation was necessary because the variables are measured on different scales, and PCA is sensitive to differences in variance across variables.

PCA was then applied to the standardised dataset to transform the original variables into a smaller number of uncorrelated principal components. These new variables are linear combinations of the original features and are ordered according to the amount of variance they explain.

3.3 Variance Explained and Number of Components Retained

Figure 5: Scree plot of variance explained by principal components



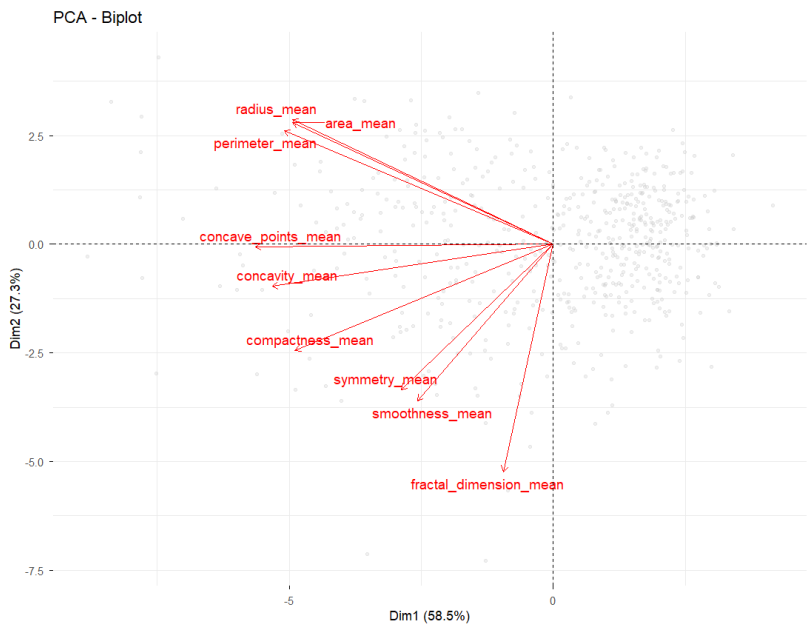
The scree plot (Figure 5) shows that the first principal component explains 58.5% of the total variance, while the second explains 27.3%. Together, the first two principal components account for approximately 85.8% of the total variance. A clear elbow is observed after the second component, indicating diminishing returns from additional components. Therefore, the first two principal components were retained for further analysis, as they provide a substantial reduction in dimensionality while preserving most of the information in the dataset.

3.4 Interpretation of the Principal Components

The loadings indicate that the first principal component (PC1) is mainly associated with radius_mean, perimeter_mean, area_mean, concavity_mean, and concave_points_mean. This suggests that PC1 captures variation related to overall nuclear size and boundary irregularity.

The second principal component (PC2) is more strongly associated with smoothness_mean, symmetry_mean, and fractal_dimension_mean. PC2 therefore appears to represent variation in nuclear texture and structural complexity.

Figure 6: PCA biplot



The PCA biplot supports this interpretation. Variables related to nuclear size are strongly aligned, confirming their joint contribution to PC1. In contrast, variables related to texture and complexity are more closely aligned with PC2, indicating that the two components represent distinct aspects of nuclear morphology.

4. Cluster Analysis

4.1 Motivation

Clustering is appropriate in this context as the dataset does not contain predefined labels. The aim is to identify natural groupings of observations based on similarities in cell nuclei characteristics, which may reflect underlying structural differences between types of cancer cells.

4.2 Choice of Methods

Two clustering methods were selected: K-means and Gaussian Mixture Models (GMM).

K-means was chosen as a baseline method due to its simplicity and effectiveness in identifying compact, well-separated clusters based on Euclidean distance. However, it assumes clusters are spherical and of similar variance.

GMM was selected as a more flexible, model-based approach. It assumes that the data arises from a mixture of Gaussian distributions, allowing clusters to have different shapes, sizes, and orientations. This makes it more suitable when clusters are not spherical. In addition, GMM provides probabilistic cluster assignments, enabling the assessment of classification uncertainty.

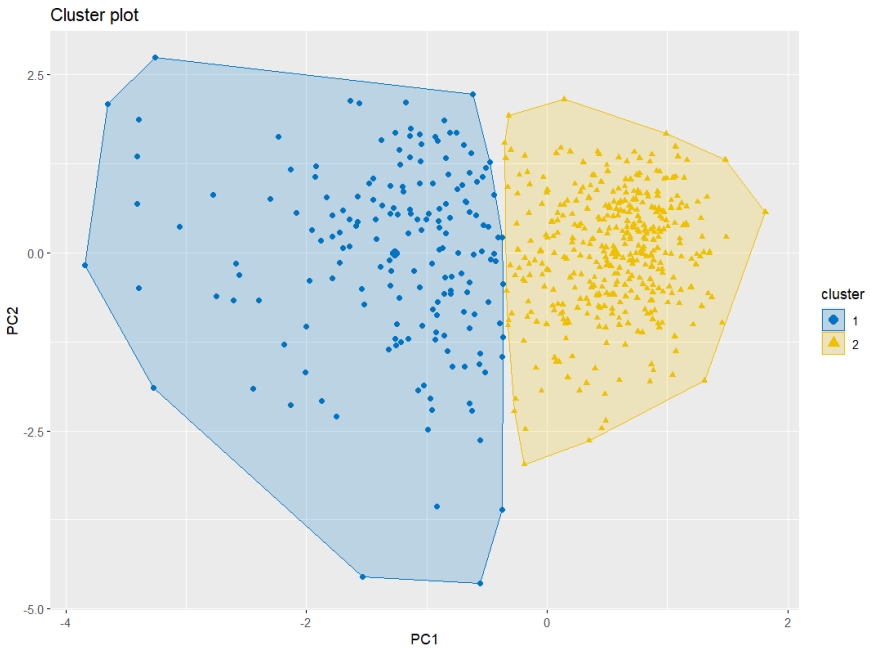
Clustering was performed on the first two principal components, which explain approximately 85% of the total variance. This reduces dimensionality and multicollinearity while retaining most of the relevant information.

4.3 Results

K-means Clustering

The optimal number of clusters was determined using both the elbow method and silhouette analysis, which consistently indicated $k = 2$.

Figure 7: K-means clustering results



The K-means clustering reveals two distinct groups separated primarily along the first principal component. One cluster corresponds to lower values of PC1, while the other corresponds to higher values.

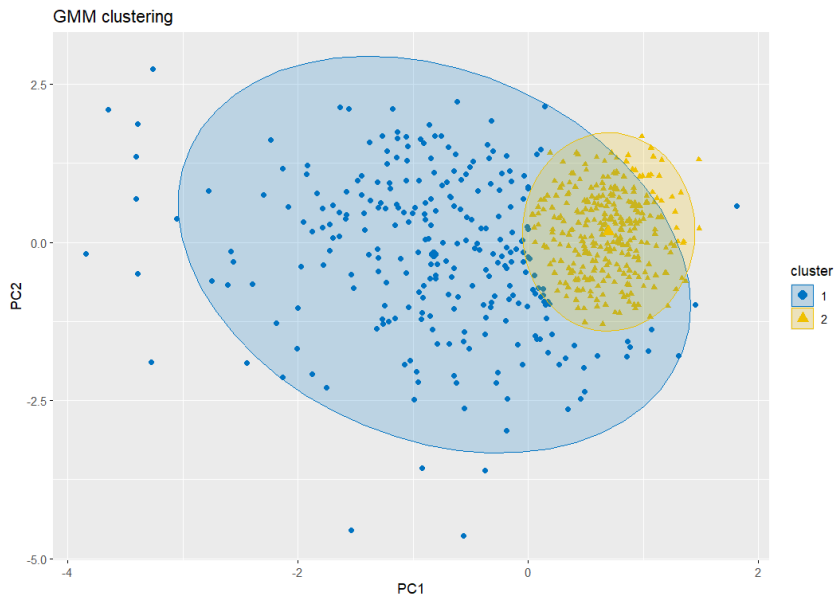
The separation between clusters occurs primarily along PC1, which represents nuclear size and boundary irregularity, indicating that these features are the main drivers of the clustering structure.

Examining the mean values of the original variables shows that one cluster is associated with smaller values of radius, perimeter, area, and concavity, indicating smaller and smoother nuclei. The second cluster exhibits substantially larger values of these variables, suggesting larger and more irregular nuclei.

Gaussian Mixture Model

The GMM was fitted using the EM algorithm, and the optimal number of clusters and covariance structure were selected using the Bayesian Information Criterion (BIC). The BIC table (appendix 2) indicates that a two-component model provides the best fit.

Figure 8: GMM clustering results



The GMM results also identify two clusters, consistent with K-means. However, the clusters (VEV) are elliptical rather than spherical, and some overlap is observed between them. This is reflected in the uncertainty plot, which shows that most observations are confidently classified, while a small number lie near the boundary between clusters.

Most observations are assigned with low uncertainty, indicating that the majority of classifications are reliable, although some overlap exists near cluster boundaries.

The overlap suggests that some observations exhibit intermediate characteristics, making them difficult to assign definitively to a single cluster.

Although the Gaussian assumption is not strictly satisfied, the approximately elliptical cluster shapes and the BIC criterion suggest that the GMM provides a reasonable approximation of the data structure.

For both methods, the clusters are reasonably balanced in size, indicating that the identified grouping is not driven by a small subset of observations.

4.4 Comparison of Methods

Both K-means and GMM identify a consistent two-cluster structure, indicating that the underlying grouping of the data is robust.

K-means achieves a higher average silhouette score (0.49), suggesting stronger separation between clusters under a distance-based metric. In contrast, GMM achieves a lower silhouette score (0.40) but provides a more flexible representation by allowing elliptical clusters with significant overlap and modelling uncertainty in cluster assignment.

The lower silhouette score for GMM does not necessarily indicate poorer performance, as GMM allows overlapping clusters and soft assignments rather than enforcing strict boundaries.

Overall, while GMM captures the cluster shape more realistically, K-means provides clearer separation.

5. Discussion and Conclusion

The analysis identified two distinct groups of observations, primarily separated by features related to nuclear size and shape irregularity. One cluster is characterised by larger and more irregular nuclei, while the other consists of smaller and smoother nuclei. This indicates the presence of two distinct structural patterns in the data, which are primarily driven by differences in nuclear size and structural irregularity.

The consistency of results across both K-means and Gaussian Mixture Models suggests that the cluster structure is robust. K-means provided clearer separation between clusters, as reflected by higher silhouette scores, while GMM offered a more flexible representation by allowing elliptical cluster shapes and modelling uncertainty in classification.

The results may vary under different methodological choices. For instance, retaining additional principal components could capture more variability but may introduce noise and reduce interpretability. Similarly, alternative clustering methods, such as hierarchical or density-based approaches, might reveal different structures. Density-based methods could capture non-linear cluster boundaries that are not well represented by distance-based approaches, while hierarchical clustering could reveal a nested structure within the data, potentially identifying subgroups rather than a single flat partition.

This analysis also has limitations. Mean imputation may reduce variability in the data, and PCA involves some loss of information due to dimensionality reduction. In addition, the clustering is performed without labelled outcomes, meaning that the interpretation of clusters cannot be directly validated. Also, clustering results may be sensitive to the choice of features and preprocessing steps, which could influence the identified structure.

Several questions remain open. In particular, it remains unclear whether the identified clusters correspond directly to clinically meaningful categories, such as benign and malignant tumours, or whether they capture more subtle variations in tumour characteristics. Further analysis incorporating labelled data would be required to confirm this relationship. In addition, it is unclear how much information was lost by removing the texture_mean variable, which contained a high proportion of missing values.

Appendix 1: R code:

```r
library(naniar)
library(visdat)
library(mice)
library(ggplot2)
library(reshape2)
library(corrplot)
library(factoextra)
library(dplyr)
library(mclust)
library(cluster)
df <- read.csv("C:/Users/cheng/Desktop/durham mds/data visualisation/breastcancer.csv")

# remove the X name column
df <- df[,-1]

str(df)
head(df)
summary(df)

# dealing with missing data
gg_miss_var(df, show_pct = TRUE) # about 50% of texture_mean and 13% of smoothness_mean missing
vis_dat(df)
md.pattern(df)

df$texture_mean <- NULL # removing column texture_mean
df$smoothness_mean[is.na(df$smoothness_mean)] <- mean(df$smoothness_mean, na.rm = TRUE) # mean imputation
# Mean imputation is adopted. This approach is simple, preserves the overall distribution of the variable, and is appropriate for downstream techniques such as PCA and clustering.
# More complex methods such as regression or multiple imputation were not considered necessary given the relatively small proportion of missing values.
# While mean imputation may reduce variance, this limitation is not critical in this context as the analysis focuses on structure rather than statistical inference.

# histogram
df_long <- melt(df)
ggplot(df_long, aes(x = value)) +
  geom_histogram(
    bins = 30,
    fill = "steelblue",
    color = "white",
    alpha = 0.9
  ) +
```

```r
  facet_wrap(~variable, scales = "free", ncol = 3) +
  theme_minimal() +
  theme(
    strip.text = element_text(size = 10, face = "bold"),
    plot.title = element_text(size = 14, face = "bold"),
    axis.text = element_text(size = 8)
  ) +
  labs(
    title = "Distribution of Variables",
    x = "Value",
    y = "Frequency"
  )

# kde
ggplot(df_long, aes(x = value)) +
  geom_histogram(aes(y = ..density..),
                 bins = 30,
                 fill = "steelblue",
                 color = "white",
                 alpha = 0.7) +
  geom_density(color = "red", size = 0.6) +
  facet_wrap(~variable, scales = "free", ncol = 3) +
  theme_minimal() +   theme(
    strip.text = element_text(size = 10, face = "bold"),
    plot.title = element_text(size = 14, face = "bold"),
    axis.text = element_text(size = 8)
  ) +
  labs(
    title = "Distribution of Variables",
    x = "Value",
    y = "Frequency"
  )
# variables such as perimeter_mean, area_mean, compactness_mean, concavity_mean etc right skewed
# radius_mean(slight right skewed), symmetry, smoothness(might be due to mean imputation) more symmetric

# boxplot
ggplot(df_long, aes(x = variable, y = value, fill = variable)) +
  geom_boxplot(alpha = 0.8, outlier.colour = "black") +
```

```r
  scale_fill_brewer(palette = "Set2") +
  theme_classic() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(
    title = "Boxplot of Variables",
    x = "",
    y = "Value"
  ) +
  guides(fill = "none")     # hard to see, might need standardisation

df_scaled <- scale(df)
df_scaled <- as.data.frame(df_scaled)
df_scaled_long <- melt(df_scaled)
ggplot(df_scaled_long, aes(x = variable, y = value, fill = variable)) +
  geom_boxplot() +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(
    title = "Boxplot of Standardised Variables",
    x = "",
    y = "Scaled Values"
  ) +
  guides(fill = "none")  # outliers present in all variables

# correlation
pairs(df) # messy to look it, not used

round(cor(df),3)
corrplot(cor(df), cl.cex=0.8, tl.col="black", tl.cex=0.9, tl.srt=45)

S
# pca

# before coding, mean imputation before might slightly reduce variance and weaken correlation. but since its only 13% it should be fine

pca <-prcomp(df_scaled)
summary(pca)
fviz_eig(pca, addlabels = TRUE) # scree plot
fviz_pca_ind(pca, label = "none") # spread mainly along pc1, no clear cluster, dense central region, some extreme points
```

```r
pca$rotation
# PC1 is mainly driven by radius_mean, perimeter_mean, area_mean, concavity_mean, and concave_points_mean
# → represents overall nuclear size and boundary irregularity
# PC2 is influenced by smoothness_mean, symmetry_mean, and fractal_dimension_mean
# → represents variation in nuclear texture and structural complexity

fviz_pca_biplot(pca,
                label = "var",
                col.var = "red",
                col.ind = "grey80",
                alpha.ind = 0.3,
                pointsize = 1,
                repel = TRUE)
# Biplot shows that radius, perimeter, and area are strongly aligned,
# indicating high correlation and contribution to overall nuclear size (PC1).
# Concavity and concave_points follow a similar direction (boundary irregularity),
# while smoothness, symmetry, and fractal_dimension relate more to PC2(texture/complexity).


# k-mean

df_pca <- pca$x[,1:2]
fviz_nbclust(df_pca, kmeans, method = "wss") +
  ggtitle("Elbow Method for Optimal k")
fviz_nbclust(df_pca, kmeans, method = "silhouette")
# pick k = 2
set.seed(123)
km2 <- kmeans(df_pca, centers = 2, nstart = 25)
fviz_cluster(km2, data = df_pca, geom = "point", ellipse.type = "convex", palette="jco")
km2

# Add cluster labels back to original data
df$km_cluster <- km2$cluster

# Examine cluster characteristics
aggregate(df, by = list(cluster = df$km_cluster), mean)
```

```r

# gmm
gmm_res <- Mclust(df_pca)
summary(gmm_res)

gmm_cluster <- gmm_res$classification
table(gmm_cluster)

fviz_cluster(list(data = df_pca, cluster = gmm_cluster),
             geom = "point",
             ellipse.type = "norm",
             palette = "jco",
             main = "GMM clustering")

plot(gmm_res, what = "uncertainty")
summary(gmm_res$uncertainty)
plot(gmm_res, what="classification")
plot(gmm_res, what = "density")
plot(gmm_res, what = "BIC")
gmm_res$BIC

df$gmm_cluster <- gmm_cluster
aggregate(df, by = list(cluster = df$gmm_cluster), mean)


# silhouette comparison of 2 model
sil_km <- silhouette(km2$cluster, dist(df_pca))
mean(sil_km[,3])
sil_gmm <- silhouette(gmm_cluster, dist(df_pca))
mean(sil_gmm[,3])

# Cluster sizes comparison
table(km2$cluster)
table(gmm_cluster)
```

Appendix 2: R Output

1. Silhouette scores:

```
> # silhouette comparison of 2 model
> sil_km <- silhouette(km2$cluster, dist(df_pca))
> mean(sil_km[,3])
[1] 0.4888043
> sil_gmm <- silhouette(gmm_cluster, dist(df_pca))
> mean(sil_gmm[,3])
[1] 0.3983306
>
```

2. Cluster size:

```
> # Cluster sizes comparison
> table(km2$cluster)

  1   2
175 394
> table(gmm_cluster)
gmm_cluster
  1   2
263 306
```

3. GMM uncertainty

```
> summary(gmm_res$uncertainty)
      Min.   1st Qu.    Median      Mean   3rd Qu.      Max.
 0.0000000 0.0005422 0.0617611 0.1021150 0.1396162 0.4903667
```

4. GMM BIC table

```
> gmm_res$BIC
Bayesian Information Criterion (BIC):
        EII       VII       EEI       VEI       EVI       VVI       EEE       VEE       EVE
1 -4784.810 -4784.810 -4710.579 -4710.579 -4710.579 -4710.579 -4716.922 -4716.922 -4716.922
2 -4626.854 -4485.120 -4729.959 -4488.540 -4637.356 -4476.316 -4618.393 -4492.820 -4619.471
3 -4578.908 -4479.320 -4583.131 -4484.540 -4576.783 -4497.265 -4637.431 -4484.987 -4615.845
4 -4597.972 -4483.412 -4602.600 -4489.732 -4601.792 -4495.733 -4607.584 -4490.893 -4608.112
5 -4616.999 -4489.575 -4621.631 -4495.686 -4626.872 -4541.165 -4626.601 -4512.395 -4633.173
6 -4560.848 -4514.500 -4632.419 -4520.921 -4595.720 -4549.072 -4556.724 -4526.402 -4557.665
7 -4580.049 -4536.991 -4557.324 -4538.465 -4551.698 -4576.167 -4576.307 -4548.236 -4586.302
8 -4561.861 -4554.188 -4545.028 -4562.856 -4570.638 -4606.086 -4550.931 -4569.011 -4578.916
9 -4553.981 -4577.483 -4540.516 -4588.022 -4586.107 -4627.599 -4546.382 -4591.734 -4616.750
        VVE       EEV       VEV       EVV       VVV
1 -4716.922 -4716.922 -4716.922 -4716.922 -4716.922
2 -4478.768 -4614.652 -4474.847 -4619.309 -4480.512
3 -4497.199 -4584.423 -4496.639 -4627.692 -4508.421
4 -4501.269 -4529.707 -4495.545 -4527.137 -4506.151
5 -4511.766 -4609.920 -4533.781 -4542.785 -4537.247
6 -4542.537 -4591.790 -4548.633 -4570.947 -4570.624
7 -4571.086 -4559.775 -4569.257 -4603.138 -4594.371
8 -4600.115 -4584.696 -4596.607 -4617.160 -4626.062
9 -4625.953 -4595.345 -4623.608 -4609.637 -4658.117

Top 3 models based on the BIC criterion:
    VEV,2     VVI,2     VVE,2
-4474.847 -4476.316 -4478.768
>
```